2025-10-08 Lecture

MH1200 Linear Algebra I

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1. The First Elementary Property

Theorem Determinant of a Matrix with a Row or Column of Zeroes — Theorem 1

The determinant of a matrix A with a row or column of zeroes is zero.

Proof Determinant of a Matrix with a Row or Column of Zeroes — Proof 2

Direct computation of the determinant along the row or column of zeroes results in zero.

2. The Second Elementary Property

TODO

3. The Third Elementary Property

Theorem Determinant Respects Linearity of Rows — Theorem 3

Let:

$$A = \begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & b\vec{B} + c\vec{C} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}$$

Then,

$$\det(A) = b \det\left(\begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{B} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}\right) + c \det\left(\begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{C} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}\right)$$

As needed. ■

Proof Determinant Respects Linearity of Rows

Proof 4

Direct computation of the determinant of A using the ith row gives:

$$\begin{aligned}
\det(A) &= \det\left(\begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & b\vec{B} + c\vec{C} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}\right) \\
&= \sum_{j=1}^{n} \left(b\vec{B}_j + c\vec{C}_j\right)(-1)^{i+j} \det\left(M_{i,j}\right) \\
&= b\sum_{j=1}^{n} \vec{B}_j(-1)^{i+j} \det\left(M_{i,j}\right) + c\sum_{j=1}^{n} \vec{C}_j(-1)^{i+j} \det\left(M_{i,j}\right) \\
&= b\det\left(\begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{B} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}\right) + c\det\left(\begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{C} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}\right)
\end{aligned}$$

As needed. ■

Corollary Determinants Respect Linear Combinations on Columns

Corollary 5

Determinants also respect linearity on specific columns. This is a direct consequence of the previous theorem and the invariance of the determinant under transposition of a matrix.

4. The Fourth Elementary Property

Theorem Repeated Row or Column Implies Zero Determinant

Theorem 6

Let A be an $n \times n$ matrix. If A has a repeated row or a repeated column, its determinant is zero.

Proof Repeated Row or Column Implies Zero Determinant

Proof 7

The proof is by mathematical induction.

First, let us consider the base case of 2×2 matrices.

Then, the matrix is of the following form:

$$\begin{pmatrix} a & b \\ a & b \end{pmatrix}$$

We note that:

$$\det\left(\begin{pmatrix} a & b \\ a & b \end{pmatrix}\right) = ab - ab = 0$$

This proves the base case.

Now, suppose that all $(n-1) \times (n-1)$ matrices (for $n \geq 3$) with a repeated row has determinant zero. We want to show that all $n \times n$ matrices with a repeated row has determinant zero.

Suppose that A is some particular but arbitrarily chosen matrix such that it has one repeated row. Since there are repeated rows, there exist rows j, k such that $j \neq k$ and $\overrightarrow{R_j} = \overrightarrow{R_k}$.

Since there are at least 3 rows in the $n \times n$ matrix, at least one of these rows is neither j nor k. Let $i \neq j \wedge i \neq k$.

Direct computation of the determinant A along row i gives:

$$\det(A) = \sum_{l=1}^{n} A_{i,l}(-1)^{i+l} \det\left(M_{i,l}\right)$$

We note that, since rows j and k are not eliminated in any of the minor matrices $M_{i,l}$, all minor matrices also include a repeated row. Furthermore, $\forall l \in [1, n] \cap \mathbb{Z}, M_{i,l} \in \boldsymbol{M}_{(n-1)\times(n-1)}(\mathbb{R})$. By the induction hypothesis, $\det\left(M_{i,l}\right) = 0$ due to the existence of repeated rows.

Therefore,

$$\begin{aligned} \det(A) &= \sum_{l=1}^{n} A_{i,l}(-1)^{i+l} \det\left(M_{i,l}\right) \\ &= \sum_{l=1}^{n} A_{i,l}(-1)^{i+l} \cdot 0 \\ &= 0 \end{aligned}$$

Thus, if all $(n-1) \times (n-1)$ matrices with repeated rows have zero determinant, then all $n \times n$ matrices with repeated rows also have zero determinant.

Therefore, by the principle of mathematical induction, all members $A \in \boldsymbol{M}_{n\times n}(\mathbb{R})$ for all $n \in \mathbb{Z}_{\geq 2}$ with a repeated row have zero determinant, as needed.

Finally, any matrix A with a repeated column has a corresponding transpose A^T with a repeated row. Thus, $\det(A) = \det(A^T) = 0$. In both cases, the determinant is zero, as needed. ■

5. The Fifth Elementary Property

Theorem Row Swap Flips Determinant Sign

Theorem 8

Suppose we can obtain A' from one row swap on A. Then,

$$\det(A') = -\det(A)$$

Proof Row Swap Flips Determinant Sign

Proof 9

Notice that:

$$0=\det\begin{pmatrix}\leftarrow & \overrightarrow{R_1} & \rightarrow\\ \leftarrow & \overrightarrow{R_2} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i}+\overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i}+\overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow\end{pmatrix}$$

$$=\det\begin{pmatrix}\leftarrow & \overrightarrow{R_1} & \rightarrow\\ \leftarrow & \overrightarrow{R_2} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i}+\overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow\end{pmatrix}+\det\begin{pmatrix}\leftarrow & \overrightarrow{R_1} & \rightarrow\\ \leftarrow & \overrightarrow{R_2} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i}+\overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow\end{pmatrix}$$

$$=\det\begin{pmatrix}\leftarrow & \overrightarrow{R_1} & \rightarrow\\ \leftarrow & \overrightarrow{R_2} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow\end{pmatrix}+\det\begin{pmatrix}\leftarrow & \overrightarrow{R_1} & \rightarrow\\ \leftarrow & \overrightarrow{R_2} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow\end{pmatrix}$$

$$+\det\begin{pmatrix}\leftarrow & \overrightarrow{R_1} & \rightarrow\\ \leftarrow & \overrightarrow{R_2} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow\end{pmatrix}+\det\begin{pmatrix}\leftarrow & \overrightarrow{R_1} & \rightarrow\\ \leftarrow & \overrightarrow{R_2} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow\\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow\end{pmatrix}$$

$$= \det \begin{pmatrix} \begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix} \end{pmatrix} + 0$$

$$+0 + \det \begin{pmatrix} \begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix} \end{pmatrix}$$

$$= \det \begin{pmatrix} \begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix} \end{pmatrix} + \det \begin{pmatrix} \begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix} \end{pmatrix}$$

Thus,

$$\det \begin{pmatrix} \begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix} \end{pmatrix} + \det \begin{pmatrix} \begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix} \end{pmatrix} = 0$$

Which implies,

$$\det\left(\begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}\right) = -\det\left(\begin{pmatrix} \leftarrow & \overrightarrow{R_1} & \rightarrow \\ \leftarrow & \overrightarrow{R_2} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_i} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_j} & \rightarrow \\ & \vdots & \\ \leftarrow & \overrightarrow{R_n} & \rightarrow \end{pmatrix}\right)$$

For any choice of $\overrightarrow{R_i}$, as needed. ■

6. The Sixth Elementary Property

Theorem Determinants of Lower and Upper Triangular Matrices — Theorem 12

The determinant of any lower or upper triangular matrix A is the product of its diagonal entries.

Proof Determinants of Lower and Upper Triangular Matrices — Proof 13

We will first prove this property on lower triangular matrices.

The proof is by induction.

First, let us show that the determinant of all 1×1 lower triangular matrices is the product of its diagonal entries.

Clearly,

$$\det(A) = A_{1,1}$$

Which is the product of all diagonal entries of the matrix, as needed.

Now, we will show that if the determinant of all $(n-1) \times (n-1)$ lower triangular matrices is $\prod_{i=1}^{n-1} A_{i,i}$, then the determinant of all $n \times n$ lower triangular matrices is $\prod_{i=1}^{n} A_{i,i}$.

Suppose $A \in \boldsymbol{M}_{n\times n}(\mathbb{R})$ is some particular but arbitrarily chosen matrix that is lower triangular.

We will compute the determinant using the top row of A.

Notice that the only nonzero entry on the top row. TODO

7. The Geometric Interpretation of a Determinant

Take two vectors $\vec{v}, \vec{w} \in \mathbb{R}$. Let P be the parallelogram:

$$P = \{a\vec{v} + b\vec{w} \mid a \in [0,1] \wedge b \in [0,1]\}$$

Then,

$$\text{Area}(P) = \left|\det\left(\begin{pmatrix}\uparrow & \uparrow \\ \vec{v} & \vec{w} \\ \downarrow & \downarrow\end{pmatrix}\right)\right|$$

In 3d, the determinant is the volume of the corresponding parallelepiped up to multiplication by 1 or -1.

8. An Alternative Definition of Determinant Based on Permutations

Theorem Equivalence of the Permutation Definition of a Determinant

Theorem 14

Let π be a permutation from $\{1, 2, 3, ..., n\}$ to itself.

Then, let $\text{sgn}(\pi) = (-1)^{\text{swaps}(\pi)}$ be the sign of a permutation, where $\text{swaps}(\pi)$ is the number of swaps to get from the indentity permutation to π.

Finally, the determinant of A is given by:

$$\det(A) := \sum_{\pi \in S_n} \left(\text{sgn}(\pi) \prod_{i=1}^{n} a_{i,\pi(i)} \right)$$